Spatiotemporal Oxygen Concentration Analysis

MATH 2130: Final project

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Main Research Area: Spatiotemporal Trends in Oxygen Concentration in the Gulf of Saint Lawrence: Short-term and Long-term Variability

Research questions:

Temporal Variability -

- How does oxygen concentration change over daily, seasonal, and long-term scales? and What kind of patterns emerge?

Spatial Variability -

- How do oxygen levels vary across different regions and water depth, are there areas where oxygen depletion is more pronounced?

Environmental Influences -

- What kinds of relationships between oxygen concentration and other environmental variables (temperature, salinity, density) are present

Methodology:

Data Collection

The data for this analysis was obtained from the “Analysis of historical environmental data collected by ocean gliders in the Gulf of Saint Lawrence”, this data was provided by the Statistical Society of Canada. Our team selected 10 out of the 15 available data sets, this was decided as there was some overlap among the data. The datasets utilized in this analysis contained historical ocean variable measurements collected through two primary methods: glider-based deployments and ship-based sampling. The glider data provided measurements of temperature, salinity, density, and oxygen concentration, while ship-based data included salinity and best_oxygen measurements from various depths.

Data Cleaning

To start preparing the data to be explored we combined all glider data into one CSV file named ‘Glider_df’. New columns were created to help facilitate analysis including depth, zone, and low_oxygen. Any missing values rows were dropped and the Glider data was grouped by (YYYY/MM/DD) which will make it easier to visualize and separate oxygen concentration over time. Additionally The bottle data underwent similar cleaning, which involved filtering out a lot of unrelated data, the timestamps were parsed and formatted as (YYYY/MM/DD) to maintain consistency across all datasets. Zone column uses a 50km grind for zoning. Due to dataset size, we randomly sampled 30 rows from each day and created a new dataframe using those samples for analysis.

Modeling and Validation:

Linear Regression:

We initially trained a linear regression model using the glider sample dataset to explore the relationship between oxygen concentration and other relevant variables. After normalization, the model returned a MSE of 538.4873, an R^2 of ~0.92, and an AIC of 23200.67 (1).

However, after creating a correlation heatmap, it showed signs of multicollinearity. (2) To combat this, we employed ridge regression along with 5-fold cross validation to find the best regularization parameter value based on mean squared error, as well as outputting its respective R^2 and AIC. We chose the lambda value based on the lowest average MSE across different lambda values, ranging from 0.1 to 100, which gave us an optimal lambda value of 0.1.

This cross validated ridge regression model gave us a MSE of 532.4732, $R^2 = 0.9200$ and AIC = 18525.7784 (3).Notice that R^2 stayed the same, MSE had a slight decrease, and AIC had a substantial decrease from 23200.67. These improvements in evaluation metrics all indicate that the ridge regression model is a better fit for our data than standard linear regression. The residuals were then plotted, and while slightly skewed, it showed similar trends to that of the normal distribution, indicating noticeable linear behaviour in our data. (4)

Random Forest:

To improve detection of low oxygen events, we trained a Random Forest classifier using spatial, temporal, and environmental features including depth, salinity, and sea water density. The model was configured with “class_weight='balanced'” to handle class imbalance. It achieved 99.7% accuracy, 0.88 precision, and 0.70 recall for the minority (low oxygen) class—making it the most effective model overall. Feature importance analysis showed depth and density as the most influential factors. Although Random Forest is less interpretable than regression models, its predictive performance was significantly stronger.

KNN Matching:

To validate glider data against the high quality bottle data, we implemented a KNN matching approach using spatial and temporal proximity. Each bottle measurement was paired with the closest glider reading based on latitude, longitude and timestamp. Initially this matching failed because all the bottle timestamps were misread as 1970-01-01. After fixing this issue, the matching succeeded, producing over 70 unique glider oxygen values (compared to the previous 1). Despite matching based on latitude, longitude and date we still observed substantial discrepancies in oxygen values. This mismatch reflects real environmental variability.

Improving Regression Models and Visual Analysis:

After using linear regression on the bottle data, we found the value of R squared was roughly 0.026. Therefore, we needed to enhance the model and check for any flaws. After removing standardization, the R squared increased to 0.764, a significant improvement. The results of the regression show that the oxygen concentration is highly influenced by longitude and latitude, as seen in their coefficients. As discussed above, multicollinearity in the glider data also needed to be addressed, leading to the use of L2 Regularization (Ridge regression). Meanwhile, when analyzing the glider data, we used Axes3D, among other 2D plots for trend observation. The 3D plot showed a disproportionately high area of low_oxygen, meaning any classification model deployed on that data could have deceivingly high accuracy, not indicative of the true nature of the data. The 2D plots helped visualize things like temporal trends, variable relationships, and data distribution.

Conclusion:

Low oxygen events in the Gulf of St. Lawrence are rare but occur more frequently in deeper waters, eastern longitudes, and later dates. Our linear regression models cover a large percentage of the dataset, providing reliable insights into the patterns behind these events. The regression results revealed that:

- Sea water temperature had the strongest impact. As sea water temperature increases, oxygen concentration decreases noticeably. (Warm water = less oxygen, expected)
- Longitude and latitude also had strong influence, and indicated a much higher oxygen concentration toward the north and eastward
- Time (month) showed a positive correlation with oxygen level, supporting a potential seasonal or progressive trend in our bottle data.

KNN matching revealed large variability in oxygen values. This variability is likely driven by natural environmental changes such as currents or stratification.

Drawbacks:

Class imbalance: A major challenge throughout the modeling process was the extreme class imbalance in the target variable, with low oxygen events representing less than 1% of the data. The models are sensitive to this imbalance.
Interpretability vs performance: Although Random Forest performed very well, it offered limited interpretability. This makes it harder to draw conclusions about long term trends or environmental patterns.
Environmental variability and KNN: The KNN matching method paired glider and bottle data based on proximity in space and time, but yielded mismatches in oxygen values. While this was useful for identifying environmental variability, it also highlighted limitations.

Feature exclusion and noise: Not all available features were included in every mode. For instance, salinity and density improved the Random Forest but tend to make linear models overfit. Additionally, we had to manually tune the thresholds to prevent misleading results.

Fig. 1: Initial linear model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.920
Model:                            OLS   Adj. R-squared:                  0.920
Method:                 Least Squares   F-statistic:                 2.273e+04
Date:                Thu, 10 Apr 2025   Prob (F-statistic):               0.00
Time:                        15:08:05   Log-Likelihood:                -53719.
No. Observations:               11796   AIC:                         1.075e+05
Df Residuals:                   11789   BIC:                         1.075e+05
Df Model:                           6
Covariance Type:            nonrobust
==================================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                        205.2694      0.212    969.468      0.000     204.854     205.684
latitude                       2.8623      0.288      9.925      0.000       2.297       3.428
longitude                     11.9649      0.292     41.003      0.000      11.393      12.537
sea_water_temperature        -66.4029      1.183    -56.126      0.000     -68.722     -64.084
depth                         -6.5002      1.126     -5.774      0.000      -8.707      -4.294
sea_water_practical_salinity  14.6089      3.543      4.124      0.000       7.665      21.553
sea_water_density           -104.2227      4.727    -22.048      0.000    -113.489     -94.957
==============================================================================
Omnibus:                       45.253   Durbin-Watson:                   2.015
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               48.282
Skew:                          -0.123   Prob(JB):                     3.28e-11
Kurtosis:                       3.195   Cond. No.                         53.9
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
Test MSE: 538.4873
Test R-squared: 0.9188
aic: 23200.6723
```

Figure 2: Correlation Heatmap

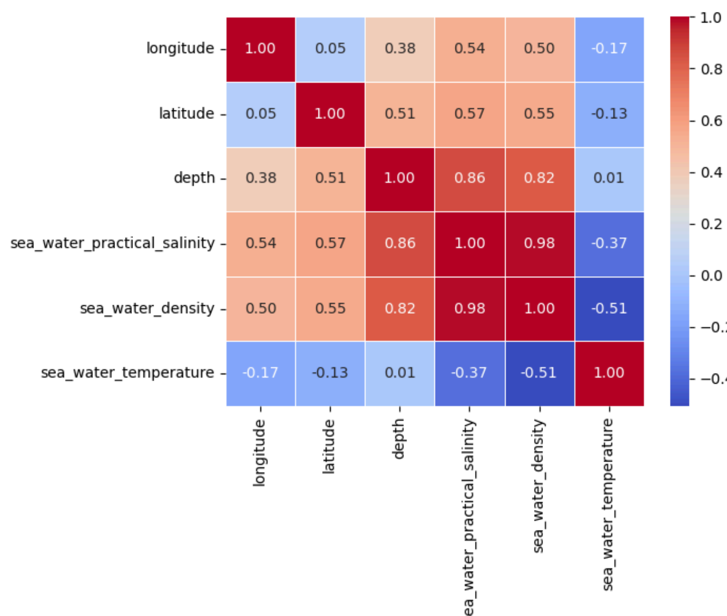

Figure 3: Ridge Regression Results

```
print(f"Best lambda by Average MSE: {best_lambda_mse:.2f} (MSE = {mean_mse[best_index]:.4f}) (R² = {mean_r2[best_index]:.4f}) (AIC = {mean_aic[best_index]:.4f}
```

```
Best lambda by Average MSE: 0.10 (MSE = 532.4732) (R² = 0.9200) (AIC = 18525.7784)
```

Figure 4: Residual QQ plot

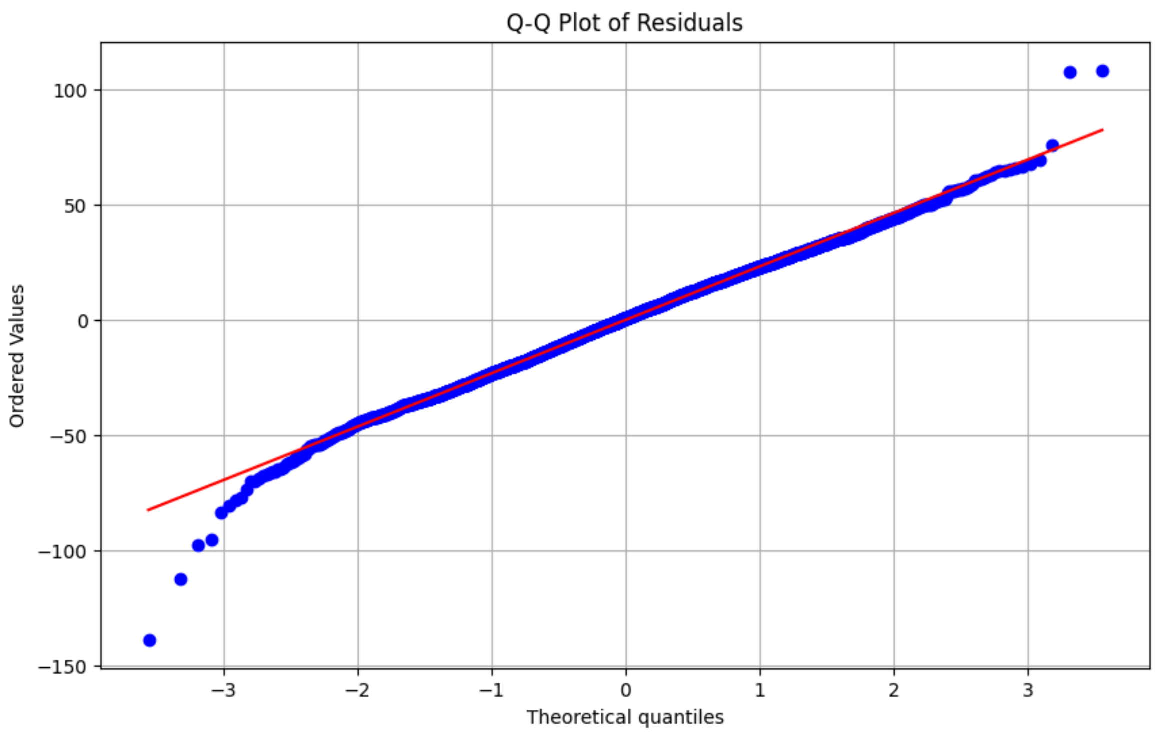

Figure 5: Axes3D

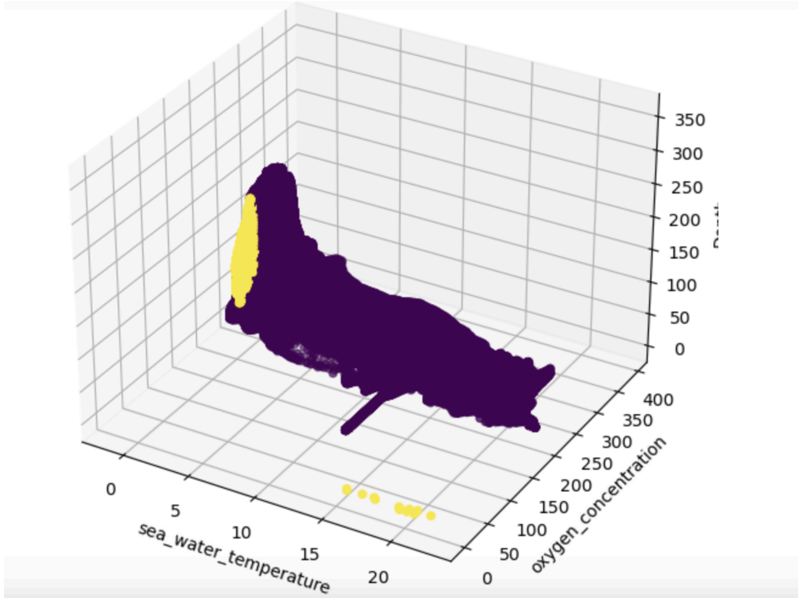

Figure 6: Better model without standardizing

OLS Regression Results

Dep. Variable:	best_Oxygen	**R-squared (uncentered):**	0.764
Model:	OLS	**Adj. R-squared (uncentered):**	0.762
Method:	Least Squares	**F-statistic:**	539.0
Date:	Mon, 07 Apr 2025	**Prob (F-statistic):**	3.16e-156
Time:	13:38:32	**Log-Likelihood:**	-3034.1
No. Observations:	503	**AIC:**	6074.
Df Residuals:	500	**BIC:**	6087.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Latitude	9.9997	2.230	4.485	0.000	5.619	14.381
Longitude	5.4378	1.748	3.111	0.002	2.004	8.872
month	3.7148	2.470	1.504	0.133	-1.137	8.567

Omnibus:	6345.831	**Durbin-Watson:**	1.136
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	55.332
Skew:	0.371	**Prob(JB):**	9.66e-13
Kurtosis:	1.554	**Cond. No.**	50.2

Figure 7: Correlation of the bottle data